

AI WORKFLOW CASE STUDY

AI-Powered Document Routing for a Regulated Audit Environment

A real operational workflow that reduced manual effort, improved classification quality, and accelerated response time.

Work Type

Anonymized

650+ audit requirements routed annually
97%+ assignment accuracy
24+ hrs manual work removed weekly

650+ items

97%+ accuracy

24+ hrs saved

Anonymized real work

Portfolio asset built for recruiter, operator, and client review.

Anonymized from real experience at a regulated financial institution. All metrics are authentic.

Santiago Garces Chavarria
Research | Strategy | AI Workflow Execution

The business problem was operational, not experimental

The workflow sat inside a recurring audit process where speed and control both mattered.

Key points

- Manual review, categorization, and routing consumed senior staff capacity every week.
- Misroutes created friction with external auditors and delayed response ownership.
- The workflow could not scale without adding headcount to a mechanical task.

Core question

How do you accelerate routing in a regulated environment without losing traceability or control?

Three bottlenecks defined the before state

The manual process was expensive because judgment and mechanics were bundled together.

Time

Every item required 15-20 minutes of reading, categorization, and forwarding.

Error risk

Inconsistent classification created escalation loops and delayed response ownership.

Scalability

Volume growth implied more headcount unless routing logic became systematic.

The workflow was redesigned as a three-layer control system

Each layer had a distinct role so automation could move fast without becoming opaque.

Classification agent

Extracts entities, assigns category, and frames the routing decision.

Routing logic

Maps the classified requirement to the correct owner through an editable matrix.

Validation layer

Routes low-confidence cases to human review and preserves an audit trail.

The technical stack stayed intentionally lightweight

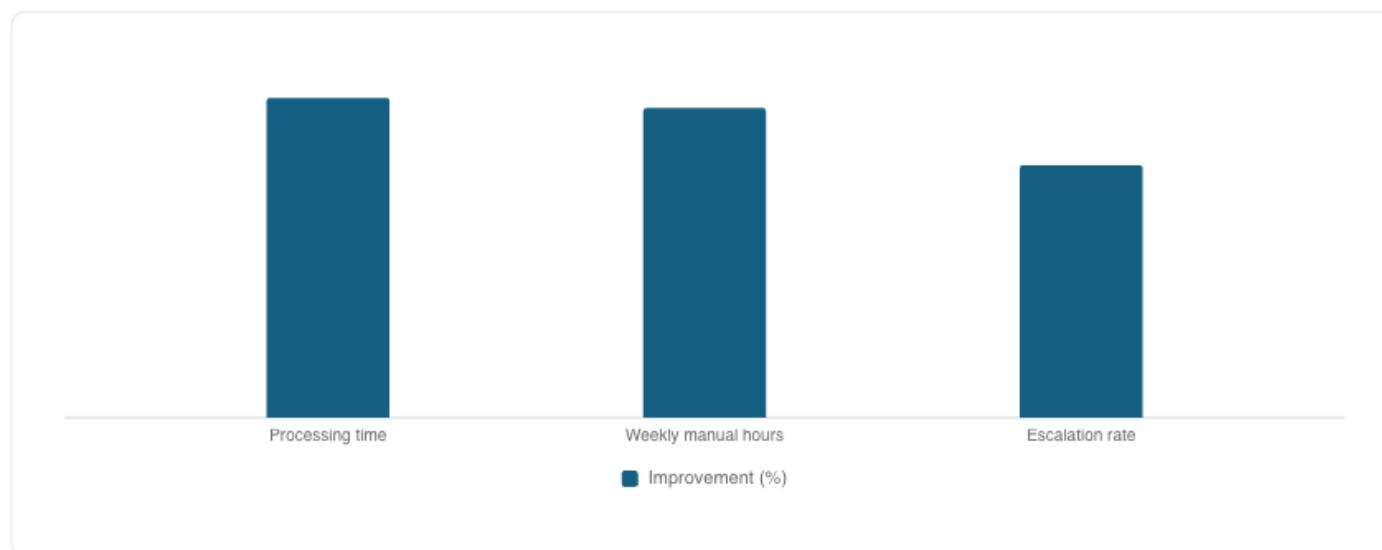
The design emphasized reliability, reviewability, and maintainable logic rather than engineering excess.

Component	Technology	Purpose
Primary LLM	Claude API	Complex requirement classification
Secondary LLM	OpenAI API	Validation and entity extraction
Orchestration	Python	Workflow logic and API integration
Routing matrix	Google Sheets	Editable control layer
Distribution	Email API	Automated routing and notification

The stack balanced speed of deployment with transparent control points.

Performance improved on the metrics that mattered most

The biggest gains came from collapsing process friction, not just reducing manual time.



Insight

Accuracy also increased from an estimated 88% to 97%+, which means the faster workflow was also more reliable.

Why the case matters beyond the audit use case

This is a template for operational intelligence work in any high-volume triage environment.

Business value

Senior staff capacity moved from sorting work to analytical work.

Control value

Faster routing came with more consistent classification and clearer ownership.

Transferability

The same pattern applies to document triage, support routing, compliance intake, and legal workflows.

Relevant for teams that need proof of execution

This case demonstrates that Santiago can connect AI tooling, decision logic, and business outcomes in environments where process quality matters.

- Suitable for workflow automation, operational intelligence, and compliance-triage discussions.
- For portfolio reviews or project discussions, email santiago.garces.blevine@gmail.com. Include the role or project type, timeline, and expected deliverable format.

Contact

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Market Research | Competitive Intelligence | AI
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Operational Intelligence Case Study